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A multi-agent system to predict the outcome of a two-round election

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ABSTRACT

Here, an agent-based model is developed to forecast the outcome of a three-party election in a two-round voting system. The vote of each agent is determined from four variables related to economic indicators, quality of public services, ethical conduct of the current administration, and political profile of the voting population. In addition, each agent can be influenced by the opinion of the nearest neighbors. Numerical simulations are performed to investigate how these variables affect the elected candidate. The proposed model is also used to reproduce the presidential elections held in Brazil in 2010 and in Uruguay in 2019.

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1. Introduction

In many democracies, direct popular elections for executive-branch positions (such as country's president and state's governor) are periodically held and their results affect the lives of virtually all citizens [1,2]. In these nations, opinion polls are conducted to forecast the winner. These polls are based on statistically representative samples; that is, the demographic and socioeconomic characteristics of the survey sample mirror those of the whole voting population [3,4]. However, even by using suitable sampling techniques, failures of pre-election polls have been reported [5,6]. Here, an alternative approach to predict the outcome of elections is introduced.

Voting dynamics can be considered as a subtopic of opinion dynamics, which has been studied, for instance, from the perspectives of complexity science, computational social science, sociophysics [7–9]. Opinion can be simplistically defined as a personal point of view about a particular subject; here, about the candidates running in an election. In theoretical works on voting dynamics, the vote is usually represented by a single scalar quantity and the models express how its value evolves over time. These works investigate, for instance, the emergence of consensus [10,11] and polarization [12,13], the influence of fanatics [14,15], the impact of nonlinear interpersonal relationships [16,17], the effects of the topological features of the network representing the social contacts [18,19]. However, in such works, the variables that explain the voting choice are not explicitly described. Here, they are. In this manuscript, the vote is determined from the perceived performance of the current administration and the political orientation of the society. Then, the influence of these variables in the outcome of a three-party election in a two-round voting system is numerically investigated. In addition, the model is employed to simulate real-world elections held in 2010 and in 2019.

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This manuscript is organized as follows. In Section 2, the proposed model is described. In Section 3, the results obtained from computer simulations are presented. In Section 4, the possible relevance of this model is discussed.

2. The model

Let the voter population be represented by a square lattice composed of $n \times n$ cells with periodic boundary conditions; that is, the left and right edges are connected and the bottom and top edges are connected too; thus, edge effects are avoided. Each cell in this toroidal lattice corresponds to a voter who is in contact with the four nearest voters (the north, east, south, and west neighbors). In the literature on cellular automata, this regular coupling topology is known as von Neumann neighborhood of unit radius [20]. Despite its simplicity, this coupling pattern has been used in studies, for instance, in game theory [21] and neurodynamics [22].

In the proposed model, the personal preference $P(k)$ of the kth individual in an election depends on four variables: $E(k)$, $S(k)$, $C(k)$, and $I(k)$. The variables E, S, and C are taken as binary and the variable I as ternary. Thus, $\{E, S, C\} \in \{0, 1\}$ and $I \in \{0, \frac{1}{2}, 1\}$. The letters E, S, C, and I respectively denote economy, service, conduct, and ideology. In the model, $E(k) = 1$ if the kth individual is satisfied with the economic indicators (such as inflation rate, interest rate, unemployment rate) and $E(k) = 0$ if is not satisfied; $S(k) = 1$ if the kth individual is pleased with the services provided by the government (such as health system, public safety, social programs, state school) and $S(k) = 0$ if is not pleased; $C(k) = 1$ if the kth individual approves the government's conduct (ethics, transparency, perceived level of corruption) and $C(k) = 0$ if does not approve. The variable $I(k)$ reflects the political ideology of the kth individual as compared to the political ideology of the current government. Assume that $I(k) = 1$ if the personal ideology coincides perfectly with the government ideology, $I(k) = 0$ if the personal ideology is completely contrary, and $I(k) = \frac{1}{2}$ if the personal ideology is between these two extremes, combining political positions of government and opposition. From these four variables, the personal preference $P(k)$ for the kth voter is given by:

$$P(k) = \frac{E(k) + S(k) + C(k) + \sigma I(k)}{3 + \sigma} \tag{1}$$

In Eq. (1), the weights of E, S, and C are equal to one and σ is the weight of I. Therefore, P is obtained from a weighted arithmetic mean of these four variables. Obviously, other weighting schemes could be considered. Note that $P = 1$ corresponds to a voter willing to elect the government candidate (because this individual is pleased with the current public administration and in line with its political agenda) and $P = 0$ corresponds to a voter who wants a drastic change.

Let Q_E be the probability of a voter being satisfied with the economic situation, Q_S the probability of a voter positively evaluating the public services, and Q_C the probability of a voter approving the current way of governing and doing politics. These three probabilities are supposed to be constants. For instance, $Q_E = 70\%$ represents a society in which, for the kth voter, $E(k) = 1$ with a 70% chance (and $E(k) = 0$ with a 30% chance). The probability distribution $Q_I = (a, b, c)$ expresses the percentages of individuals with $I = 0$, $I = \frac{1}{2}$, and $I = 1$, respectively (of course, $a + b + c = 1$). For instance, $Q_I = (\frac{1}{2}, 0, \frac{1}{2})$ describes a politically polarized society.

In this multi-agent system, the social interactions are taken into consideration in the vote variable $V(k)$, which is defined by:

$$V(k) = \frac{P(k) + q \sum_{j=1}^{\nu} P(j)}{1 + q\nu} \tag{2}$$

In Eq. (2), the non-negative parameter q linearly weights the neighborhood effect on the vote of the kth individual. Therefore, $q = 0$ denotes a society composed of self-centered individuals, and the higher the value of q, the stronger the collective way of thinking. This parameter reflects the usual influence of family, friends, and coworkers in an election. The denominator $1 + q\nu$ is a normalization factor. The index j labels the ν neighbors (here, $\nu = 4$). Notice that a linear interpersonal coupling, in which the opinion of an agent is affected by a linear combination of the opinion of its neighbors, was introduced in seminal works on opinion dynamics [23–25]. Notice also that surveys can be carried out to determine the values of the parameters q and σ to better characterize a pre-election scenario.

Suppose that three candidates called G, M, and O (representing political parties or coalitions) are running in an election in a two-round voting system. Let G be the government candidate, O the opposition candidate, and M the moderate candidate, which is between G and O in the political spectrum. In the proposed model, the vote of the kth individual in the first round is ruled by:

- vote in G if $\beta < V(k) \leq 1$,
- vote in M if $\alpha \leq V(k) \leq \beta$,
- vote in O if $0 \leq V(k) < \alpha$.

The non-negative constants α and β define the intervals of $V(k)$ corresponding to each candidate. A candidate must receive an absolute majority of votes to be declared elected. Hence, if none of the candidates gets more than 50% of the votes in the first round, then a second round with the two most voted candidates is held. In the second round, the vote of the kth individual is ruled by:

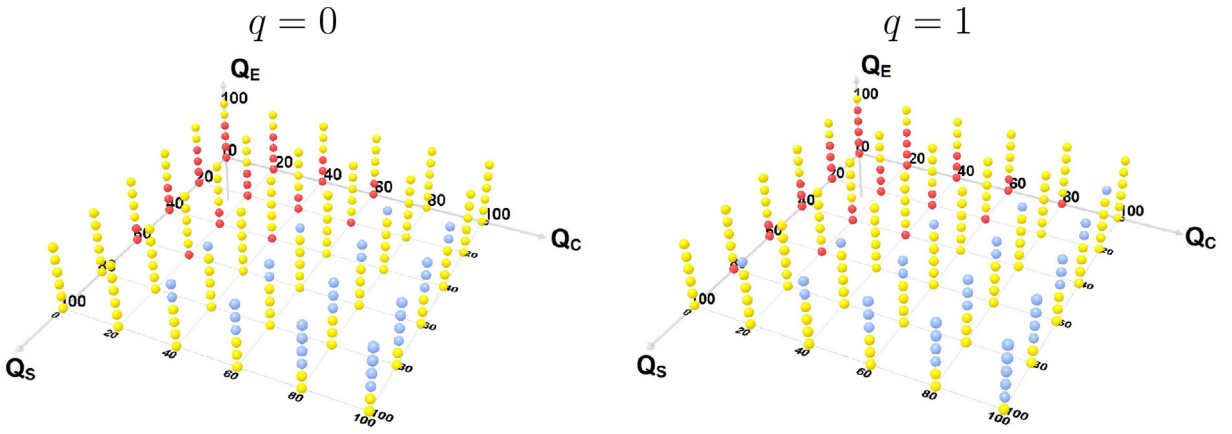


Fig. 1. Elected candidate in function of Q_E, Q_S, and Q_C in the politically balanced scenario; that is, $Q_I = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, for $q = 0$ (left) and $q = 1$ (right). Blue ball denotes the candidate G, yellow ball the candidate M, and red ball the candidate O. The other parameter values are the same as shown in Table 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

- in a second round with G and M, vote in G if $\beta < V(k) \leq 1$ and vote in M if $0 \leq V(k) \leq \beta$ (therefore, O voters choose M);
- in a second round with M and O, vote in M if $\alpha \leq V(k) \leq 1$ and vote in O if $0 \leq V(k) < \alpha$ (therefore, G voters choose M);
- in a second round with G and O, vote in G if $\frac{\alpha+\beta}{2} < V(k) \leq 1$ and vote in O if $0 \leq V(k) \leq \frac{\alpha+\beta}{2}$ (therefore, the votes in M in the first round are distributed proportionally between G and O).

These voting rules can be easily adapted for a two-party election: if the candidates are G and O, then take $\alpha = \beta$; if the candidates are G and M, then take $\alpha = 0$; if the candidates are M and O, then take $\beta = 1$.

In short: to simulate an election, the $N = n^2$ individuals have their values of *E, S, C*, and *I* randomly picked from the probability distributions Q_E, Q_S, Q_C, and Q_I, respectively, which can be inferred from official statistics. From these four variables and the ideology weight σ, the personal preference $P(k)$ is calculated and, then, by taking into account the neighborhood weight q, the vote variable $V(k)$ is computed for all individuals. The votes for each candidate are determined by the voting rules and these votes are added. The results of numerical simulations are shown in the next section.

3. Numerical results

Numerical simulations were carried out in order to evaluate the influence of the probability distributions Q_E, Q_S, Q_C, and Q_I on the outcome of a two-round election. These simulations were run by taking $n = 100$ (thus, the population is composed of $N = 10000$ voters), $\sigma = 1$ (with this choice, the weights of the four variables in $P(k)$ are equal), $\alpha = \frac{1}{3}$ and $\beta = \frac{2}{3}$ (therefore, in the voting rules, the intervals of $V(k)$ related to the three candidates have the same length). In addition, $\{Q_E, Q_S, Q_C\} = \{0, 20\%, 40\%, 60\%, 80\%, 100\%\}$. All the $6 \times 6 \times 6 = 216$ combinations of these three binary variables were simulated for $q = \{0, 1\}$ (the neighborhood weight) and for the following four distributions $Q_I = (a, b, c)$ (recall that *a, b*, and *c* are the percentages with $I = 0$, $I = \frac{1}{2}$, and $I = 1$, respectively):

- $Q_I = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$: the politically balanced scenario,
- $Q_I = (\frac{1}{2}, 0, \frac{1}{2})$: the politically polarized scenario,
- $Q_I = (\frac{3}{4}, 0, \frac{1}{4})$: the pro-opposition scenario,
- $Q_I = (\frac{1}{4}, 0, \frac{3}{4})$: the pro-government scenario.

Therefore, $216 \times 2 \times 4 = 1728$ simulations were performed. The election can either finish in the first round or require the second round. Figs. 1–4 show the elected candidate in each of these 1728 cases. In these figures, the candidate G is represented by a blue ball, M by a yellow ball, and O by a red ball. Table 1 summarizes the election results in function of q and Q_I. As expected, in these figures, G tends to win for values of Q_E, Q_S, and Q_C close to one, and O tends to win for values close to zero.

A real-world case: the presidential election held in Uruguay in 2019. In this election, there were 11 candidates [26,27] with the following political positions: 3 were left-wing oriented, 3 were right-wing oriented, and 5 were from centre parties. To adapt this electoral scenario to the model, the 11 candidates were grouped into 3 coalitions: left-wing candidates correspond to the candidate G, the right-wing candidates to the candidate O, and the centre candidates to the candidate M. In the first round, the percentages of valid votes for G, M, and O were 41.4%, 16.3%, and 42.3% [28]. In the second round between G (Daniel Martínez, left wing, 40.5% in the first round) and O (Lacalle Pou, right wing, 29.7% in the first round), the percentages were 49.4% and 50.6%; hence, O (Lacalle Pou) was elected [28].

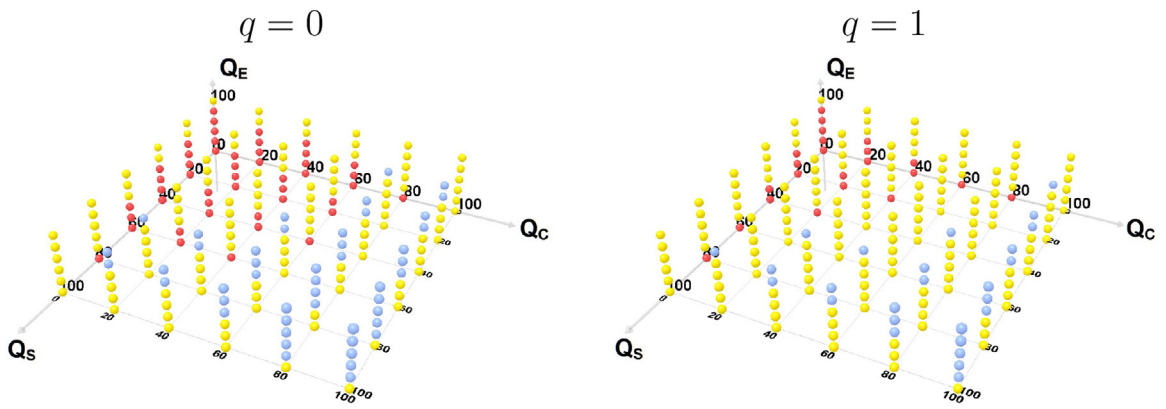


Fig. 2. Elected candidate in function of Q_E, Q_S, and Q_C in the politically polarized scenario; that is, $Q_I = (\frac{1}{2}, 0, \frac{1}{2})$, for $q = 0$ (left) and $q = 1$ (right). The other parameter values are the same as shown in Table 1.

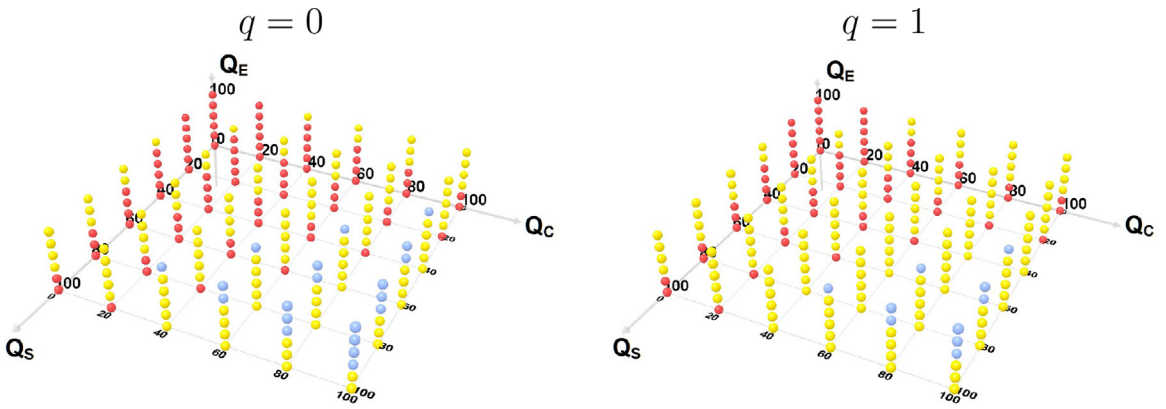


Fig. 3. Elected candidate in function of Q_E, Q_S, and Q_C in the pro-opposition scenario; that is, $Q_I = (\frac{3}{4}, 0, \frac{1}{4})$, for $q = 0$ (left) and $q = 1$ (right). The other parameter values are the same as shown in Table 1.

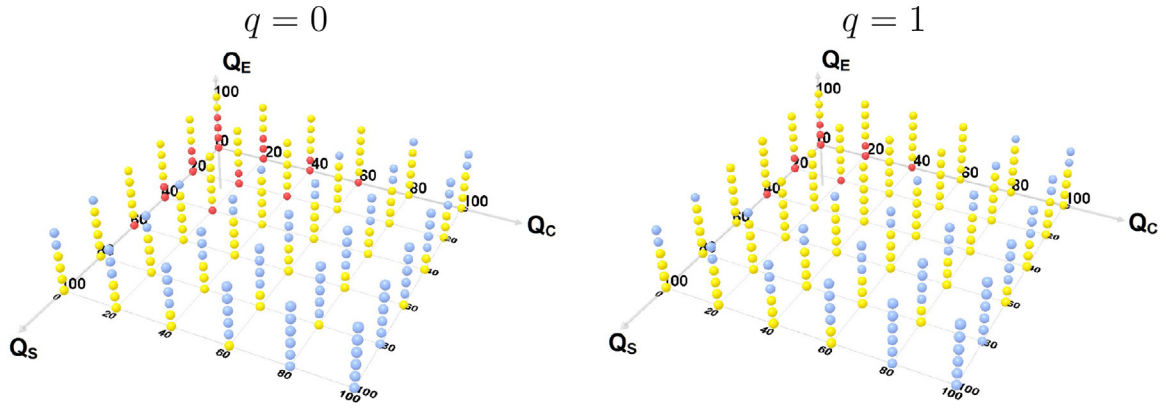


Fig. 4. Elected candidate in function of Q_E, Q_S, and Q_C in the pro-government scenario; that is, $Q_I = (\frac{1}{4}, 0, \frac{3}{4})$, for $q = 0$ (left) and $q = 1$ (right). The other parameter values are the same as shown in Table 1.

Consider that, in the pre-electoral period, $Q_E = \frac{1}{2}$ (based on the consumer confidence index [29]), $Q_S = \frac{1}{3}$ (based on the government's approval rating; public safety was a major reason for dissatisfaction [30]), and $Q_C = \frac{7}{10}$ (based on the corruption perceptions index [31]). As the Uruguayan electorate is highly polarized [30], then $Q_I = (\frac{1}{2}, 0, \frac{1}{2})$. A local search around the numbers for α, β, σ, and q used in Table 1 and Figs. 1–4 was performed, in order to fit the model. From this search, convenient values for these four parameters were found. Thus, by taking $\alpha = \frac{2}{5}$, $\beta = \frac{3}{5}$, $\sigma = 2$, and $q = \frac{1}{2}$, the model

Table 1
Winning percentage of each candidate in function of q and Q_I. In these simulations, $n = 100$, $\sigma = 1$, $\alpha = \frac{1}{3}$, $\beta = \frac{1}{3}$, and $\{Q_E, Q_S, Q_C\} = \{0, 20\%, 40\%, 60\%, 80\%, 100\%\}$. Each line corresponds to the sum of winnings in 216 combinations of Q_E, Q_S, and Q_C. This table summarizes the results shown in Figs. 1–4.

The politically balanced scenario			
q	G wins	M wins	O wins
0	15%	70%	15%
1	16%	68%	16%
The politically polarized scenario			
q	G wins	M wins	O wins
0	22%	56%	22%
1	12%	77%	11%
The pro-opposition scenario			
q	G wins	M wins	O wins
0	9%	53%	38%
1	4%	67%	29%
The pro-government scenario			
q	G wins	M wins	O wins
0	38%	53%	9%
1	29%	67%	4%

predicts that, in the first round, the percentages of votes for G, M, and O are 43.1%, 11.1%, and 45.8%, and the second round between G and O, the percentages of votes are 49.6% and 50.4%, which are in good agreement with the data.

Another real-world case: the presidential election held in Brazil in 2010. In this election, 3 candidates received about 99% of the valid votes in the first round [32]. In this round, the percentages of votes for G (Dilma Rousseff), M (Marina Silva), and O (José Serra) were 47%, 19%, and 33%. In the second round between G and O, the percentages were 56% and 44%; therefore, G (Dilma Rousseff) was the winner [32].

Assume that, in the pre-electoral period, $Q_E = \frac{3}{5}$ (based on the family expectation index [33]), $Q_S = \frac{9}{10}$ [34] (based on the government's approval rating; the socio-economic policy was a major reason for satisfaction [35]), and $Q_C = \frac{2}{5}$ (based on the corruption perceptions index [36]). As the previous four elections were polarized between the political parties of G and O [37], then $Q_I = (\frac{1}{2}, 0, \frac{1}{2})$. The model was tuned by taking the same values of α, β, and σ used in the Uruguayan election; only the value of q is different. Thus, for $\alpha = \frac{2}{5}$, $\beta = \frac{3}{5}$, $\sigma = 2$, and $q = \frac{1}{5}$, the model predicts that, in the first round, the percentages of votes for G, M, and O are 49%, 19%, and 32%, and the second round between G and O, the percentages of votes are 55% and 45%. Note that these numbers are in good agreement with the data.

These results are discussed in the next section.

4. Discussion and conclusion

In democratic societies, a central issue is to understand the decision-making process in political elections. In theoretical studies on voting dynamics, the voting intention is expressed by a real number; however, how this number can be obtained from real-world variables is not clarified. In fact, identifying the set of variables which would explain the voting behavior in any election is a challenge. For instance, in the 2016 United States presidential election, the relation between votes for the Democratic Party's candidate (Hillary Clinton) and the level of formal education was linear; in the 2012 election, the relation between votes for the Democratic Party's candidate (Barack Obama) and the level of formal education had a U-shaped pattern [6]. Thus, this relation changed from an election to another.

Here, voters are represented by socially interacting agents. The personal preference $P(k)$ and, consequently, the vote variable $V(k)$ are supposed to linearly depend on E, S, C, and I. The probability distributions of these variables, denoted by Q_E, Q_S, Q_C, and Q_I, can be inferred from opinion surveys conducted in the pre-electoral period of a particular election. Note that the voter's perception of the current administration determines Q_E, Q_S, and Q_C. It is relevant to stress that this perception is influenced by the traditional media and the virtual social networks, which spread true news and fake news.

Figs. 1–4 exhibit a portfolio of electoral scenarios in the parameter space $Q_E \times Q_S \times Q_C$ for two values of q and four distinct distributions Q_I. Indeed, a myriad of electoral scenarios could be simulated for other choices of q, Q_I, σ, α, and β. From the political marketing standpoint, these portfolios can be used in election campaign planning. Consider, for instance, the cases with $Q_E = Q_S = 100\%$ and $q = 0$ shown in Fig. 1. In these cases, M wins for $Q_C = \{0, 20\%\}$ and G wins for $Q_C = \{40\%, 60\%, 80\%, 100\%\}$. Thus, Q_C decides the winner. Notice that a widely employed strategy in election campaigns is to spread news about the government corruption, in order to change the value of Q_C.

Today, in many democracies, the population is politically divided into two groups of similar sizes having contrasting positions [38,39]. Such a deep polarization stimulates segregation and conflicts. Fig. 2 shows that $q = 1$ can hinder the

victory of the candidates G and O, as compared to $q = 0$. In fact, in this politically polarized scenario, the higher the value of q, the higher the number of victories of the candidate M. For instance, the winning percentages of G, M, and O for $q = \frac{1}{2}$ (not shown in Table 1) are 17%, 66%, and 17%, respectively. Since the election of either G or O displeases almost half the population, voters could take into account the neighbors' opinion (that is, $q > 0$) in their voting choice. In practice, the challenge is to find M, the potential consensus candidate. In Fig. 1, in a politically balanced society, the influence of q is less relevant. Figs. 3 and 4 represent complementary political profiles; hence, the results shown in Table 1 are symmetric. In these two figures, the number of combinations of Q_E, Q_S, and Q_C in which M wins also increases with q.

The proposed model can be seen as an alternative or complement to opinion polls. By tuning the values of σ, q, α, and β, the model was capable of reproducing the winners of two presidential elections. Observe that the values of these four parameters can be characteristic of each country at a given historical moment; therefore, they can be affected by socio-cultural factors.

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References

[1] L. Diamond, M.F. Plattner (Eds.), Electoral Systems and Democracy, Johns Hopkins University Press, Baltimore, 2006.
[2] G. Passarelli, Preferential Voting Systems: Influence on Intra-Party Competition and Voting Behaviour, Palgrave Macmillan, London, 2019.
[3] N. Moon, Opinion Polls: History, Theory and Practice, Manchester University Press, Manchester, 1999.
[4] J. Bethlehem, Understanding Public Opinion Polls, CRC Press, Boca Raton, 2018.
[5] P. Sturgis, J. Kuha, N. Baker, M. Callegaro, S. Fisher, J. Green, W. Jennings, B.E. Lauderdale, P. Smith, An assessment of the causes of the errors in the 2015 UK general election opinion polls, J. R. Statist. Soc. A 181 (2018) 757–781.
[6] C. Kennedy, M. Blumenthal, S. Clement, J.D. Clinton, C. Durand, C. Franklin, K. McGeeney, L. Miringoff, K. Olson, D. Rivers, L. Saad, G.E. Witt, C. Wlezien, An evaluation of the 2016 election polls in the United States, Public Opin. Q. 82 (2018) 1–33.
[7] C. Castellano, S. Fortunato, V. Loreto, Statistical physics of social dynamics, Rev. Mod. Phys. 81 (2009) 591–646.
[8] A.V. Proskurnikov, R. Tempo, A tutorial on modeling and analysis of dynamic social networks. Part I, Annu. Rev. Control 43 (2017) 65–79.
[9] L. Mastroeni, P. Vellucci, M. Naldi, Agent-based models for opinion formation: a bibliographic survey, IEEE Access 7 (2019) 58836–58848.
[10] J.R.P. French Jr., A formal theory of social power, Psychol. Rev. 63 (1956) 181–194.
[11] M.W. Li, H. Dankowicz, Impact of temporal network structures on the speed of consensus formation in opinion dynamics, Physica A 523 (2019) 1355–1370.
[12] T. Krueger, J. Szwabiński, T. Weron, Conformity, anticonformity and polarization of opinions: insights from a mathematical model of opinion dynamics, Entropy 19 (2017) 371.
[13] V. Amelkin, F. Bullo, A.K. Singh, Polar opinion dynamics in social networks, IEEE Trans. Autom. Control 62 (2017) 5650–5665.
[14] S. Galam, F. Jacobs, The role of inflexible minorities in the breaking of democratic opinion dynamics, Physica A 381 (2007) 366–376.
[15] S.L. Wang, L.B. Rong, J.H. Wu, Bistability and multistability in opinion dynamics models, Appl. Math. Comput. 289 (2016) 388–395.
[16] R. Hegselmann, U. Krause, Opinion dynamics and bounded confidence: models, analysis and simulation, JASSS-J. Artif. Soc. S. 5 (2002) 2.
[17] A. Naimzada, M. Pireddu, Adaptive decision dynamics: bifurcations, multistability and chaos, Appl. Math. Comput. 239 (2014) 375–390.
[18] F. Bagnoli, R. Rechtman, Bifurcations in models of a society of reasonable contrarians and conformists, Phys. Rev. E 92 (2015) 042913.
[19] M.B. Ye, J. Liu, B.D.O. Anderson, C.B. Yu, T. Basar, Evolution of social power in social networks with dynamic topology, IEEE Trans. Autom. Control 63 (2018) 3793–3808.
[20] S. Wolfram, Cellular Automata and Complexity: Collected Papers, Westview Press, Boulder, 1994.
[21] L.B.V. Alves, L.H.A. Monteiro, A spatial evolutionary version of the ultimatum game as a toy model of income distribution, Commun. Nonlinear Sci. Numer. Simulat. 76 (2019) 132–137.
[22] T.G. Correale, L.H.A. Monteiro, On the dynamics of axonal membrane: ion channel as the basic unit of a deterministic model, Appl. Math. Comput. 291 (2016) 292–302.
[23] R.P. Abelson, Mathematical models of the distribution of attitudes under controversy, in: N. Frederiksen, H. Gullicksen (Eds.), Contributions to Mathematical Psychology, Holt, Rinehart & Winston, New York, 1964, pp. 142–160.
[24] M.H. DeGroot, Reaching a consensus, J. Am. Stat. Assoc. 69 (1974) 118–121.
[25] N.E. Friedkin, E.C. Johnsen, Social influence and opinions, J. Math. Sociol. 15 (1990) 193–205.
[26] Noticias de América Latina y el Caribe (NODAL), Qué se vota en Uruguay y quiñes son los candidatos, 2020, https://www.nodal.am/2019/10/que-se-vota-en-uruguay-y-quienes-son-los-candidatos/ (accessed 18 April 2020).
[27] Americas quarterly, meet the candidates: Uruguay, 2020, https://www.americasquarterly.org/content/meet-candidates-uruguay (accessed 18 April 2020).
[28] Electoral geography 2.0, Uruguay presidential election, 2019. https://www.electoralgeography.com/new/en/countries/u/uruguay/uruguay-presidential-election-2019.html (accessed 18 April 2020).
[29] Universidad Católica del Uruguay, Índice de confianza del consumidor (ICC) - junio, 2019, https://ucu.edu.uy/sites/default/files/facultad/fce/economia/ICC/2019/indice_confianza_consumidor_junio_2019.pdf (accessed 18 April 2020).
[30] E.E. Townsend-Bell, Uruguay 2018: a year of mixed signals and open questions, Rev. Cienc. Polit. 39 (2019) 367–390.
[31] Transparency International, Corruption perceptions index, 2019, https://www.transparency.org/cpi2019 (accessed 18 April 2020).
[32] L. Morais, A. Saad-Filho, Brazil beyond Lula – Forging ahead or pausing for breath? Lat. Am. Perspect. 38 (2011) 31–44.
[33] IPEA, Índice de Expectativa das Famílias, n. 3, 2010, http://ipea.gov.br/portal/images/stories/PDFs/IEF/IEF_novembro2010.pdf (accessed 18 April 2020).
[34] The Guardian, Lula era comes to an end in Brazil, 2020, https://www.theguardian.com/world/2010/dec/31/brazil-lula-era-ends (accessed 18 April 2020).
[35] J.F. Mayer, The limits of labor legislation reforms: rigidity, growth, and employment in Brazil (1995–2010), J. Polit. Lat. Am. 8 (2016) 95–127.
[36] Transparency International, Corruption perceptions index, 2010. https://www.transparency.org/cpi2010 (accessed 18 April 2020).
[37] E.U. Cervi, Spatial analysis of electoral performance of PT and PSDB between 1994 and 2014 in Brazil, América Latina Hoy – Revista de Ciencias Sociales 75 (2017) 55–74.
[38] H. Gersbach, P. Muller, O. Tejada, Costs of change and political polarization, Eur. J. Polit. Econ. 60 (2019) 101801.
[39] P.F. Ribeiro, Partidos políticos y política exterior en América latin, Revista Mexicana de Ciencias Polticas y Sociales 64 (2019) 353–393.